# Pranav Senthilkumar

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## Education

**Texas A&M University** — College Station, TX
*Bachelor of Science in Electrical Engineering, Engineering Honors* — *Expected May 2028*
- **Coursework:** C++, Python, Data Structures and Algorithms, Machine Learning, Multivariable Calculus, Differential Equations
- **Independent Study:** Data Structures and Algorithms

## Experience

**Robotics Software Engineering Intern** — August 2026 - Present
*Nietzsche Labs* — *Remote*
- Built a cross-platform Electron flasher for robot OS provisioning on Windows, macOS, and Linux, integrating with a production backend on AWS API Gateway and Lambda
- Protected host machines from destructive disk writes with 8 pre-write safety checks and target re-verification before flashes
- Automated first-boot filesystem expansion across 10 SD-card sizes (2GB–256GB), eliminating manual partition resizing with zero failures across 15 validation runs

**Undergraduate Machine Learning Researcher** — June 2026 - Present
*LIVE Lab, Texas A&M University* — *College Station, TX*
- Building an ML pipeline for mesoscale-to-microscale wind downscaling across diverse terrain types, comparing general vs. terrain-specific model architectures
- Benchmarked XGBoost, TabPFN, and GP/SVGP residual correction models across 6 weather variables for wind speed prediction
- Built a geometry pipeline resolving heights for 500+ buildings across multiple Texas towns using Microsoft's Building Footprints API, LiDAR, and OSM

**Student Researcher** — June 2024 – October 2024
*Algoverse AI Research* — *Remote*
- Aligned next-token probability distributions with human ethical judgments on morally ambiguous scenarios, calibrating LLM outputs against real-world annotated moral datasets
- Fine-tuned Llama-3, Zephyr-7B, and Mistral-7B with QLoRA, improving Dirichlet loss by up to 31% relative to base model alignment on moral reasoning tasks
- Benchmarked alignment on ethical NLP datasets (Anecdotes & Dilemmas) using cross-entropy and Dirichlet multinomial loss

## Projects

**Regdrift** | *Python, CMSIS-SVD, pytest, GitHub Actions* — June 2026 – July 2026
- Built a Python CLI and GitHub Action blocking 22 classes of firmware-breaking CMSIS-SVD changes before merge.
- Implemented deterministic diffs for inheritance, arrays, clusters, registers, fields, interrupts, and access semantics.
- Validated 330 tests across 15 SVDs from STM32, Nordic, NXP, Atmel, and RP2040 ecosystems.
- Published to PyPI and GitHub Marketplace with JSON reports, Python 3.11–3.12 CI, and automated release verification.

**2-Axis Gimbal** | *Arduino, C++, GY-87 IMU, SG90 Servos, SolidWorks* — May 2026 – Present
- Wired and programmed 2-DOF gimbal in C++, fusing GY-87 IMU data and dual servo PWM outputs for automatic stabilization
- Implemented PID control loops to correct orientation drift, tuning gains for smoother tracking
- Extending the system with YOLO-based object tracking to drive gimbal orientation from live camera input

**ThermGuard** | *Python, PyTorch, Conformal Prediction* — May 2026 – June 2026
- Modeled safe per-core temperature bounds for multi-core chips in PyTorch and used them to schedule tasks that keep workloads under thermal limits
- Trained quantile regression models and applied conformal calibration to turn raw predictions into guaranteed upper bounds that held 95% coverage against a 90% target across 5 seeds
- Validated the pipeline with an automated multi-seed evaluation harness, isolating exactly where calibration holds and where it breaks under distribution shift

**Haptic Portal** | *Python, DepthAI, OpenCV, MediaPipe* — January 2026 – May 2026
- Prototyped a real-time vision-to-haptics system with hand and depth tracking, awarded Best Pitch and $250
- Converted depth frames into a 5×5 grid of 25 normalized values, compressing depth data for low-latency control
- Applied temporal and median filtering to stabilize noisy depth readings, then routed the signal to a custom motor array

## Technical Skills

**Languages:** Python, C++, Java, JavaScript, SQL
**Machine Learning:** PyTorch, TensorFlow, XGBoost, scikit-learn, Gaussian Processes, LLM Fine-Tuning, NLP
**Web, Data & Tools:** React, Node.js, MCP, NumPy, Pandas, Git, GitHub Actions, Linux

## Publication

**Pranav Senthilkumar, et al.** "Fine-Tuning Language Models for Ethical Ambiguity: A Comparative Study of Alignment with Human Responses." NeurIPS 2024 SoLaR Workshop | arXiv:2410.07826